2.1. Inductive Method

Inductive method is based on the process of induction. It leads from concrete to abstract, particular to general and from examples to general rules. In this we first take a few examples and then generalize. Thus it is a method of constructing a formula with the help of a sufficient number of concrete examples. Induction means to provide a universal truth by showing that if it is true for a particular case, it is true for all such cases. A formula or a generalization is thus arrived at through a convincing process of reasoning and solving problems.

For example if we wish to frame the formula $(a-b)^2 = a^2 - 2ab + b^2$. Let the students actually multiply (a-b) x (a-b) and find out the product. They may then be asked to find the answers for $(p-q)^2$, $(l-m)^2$ etc., by actual multiplication. After this they may be asked to observe results and be helped to make generalization to get the required formula.

Many other examples can be quoted in support of inductive method.

$$\text{Simple Interest} = \frac{\text{Principal x rate x time}}{100}$$

$$\text{Average} = \frac{\text{Sum of terms}}{\text{Number of terms}}$$

2.1.1 Merits of Inductive Method

i. It is a psychological method and in this method the interest of the student is sustained till the end.

ii. It helps in understanding because the students come to know how a particular formula has been framed.

iii. It is a natural method of making discoveries. Majority of discoveries have been made inductively.

iv. Since it is a logical method so it suits teaching of mathematics.

v. It encourages active participation of students and thus stimulates self-effort on their part.

vi. It discourages cramming and also reduces homework.

vii. It does not burden the mind. Formula becomes easy to remember.

viii. It is based on actual observations, thinking and experimentation.

ix. This method is found to be suitable in the beginning stages. All teaching in mathematics is inductive in the beginning.

x. It helps in increasing student-teacher contact.

2.1.2 Demerits of Inductive Method

i. This method is limited in range and is not suitable for all topics. Certain complex and complicated formulae cannot be generalized in this manner.

ii. It is time consuming and laborious method.

iii. Inductive reasoning is not absolutely conclusive because the generalization made with the help of a few specific examples may not hold well in all cases.

iv. We don't complete the study of a topic simply by discovering a formula but a lot of supplementary work and practice is required for fixing the topic in learner's mind.

2.1.3. Application of Inductive Method

Its application has to be restricted and confined to understanding of rules in the early stages. This method is quite useful in lower classes where certain simple generalization is to be made. At the advanced stage it is not so useful. This method is psychological and gives very good results if followed by the deductive method.